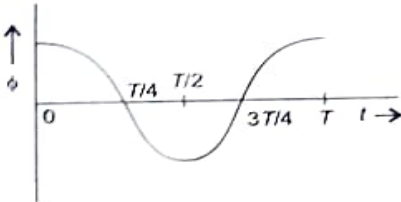
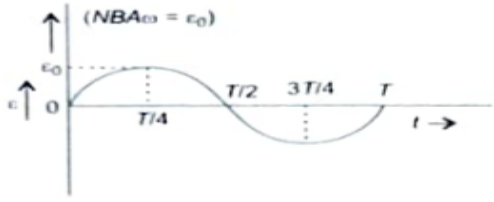
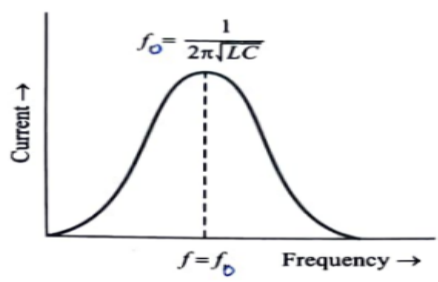


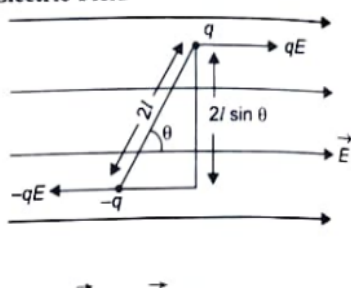
Subject-Physics
Class-XII

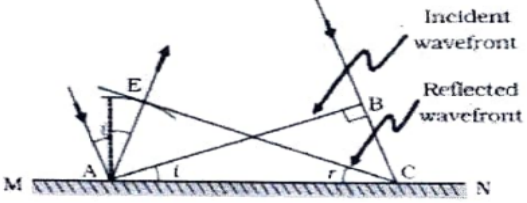
Marking Scheme		
Section-A		
1	(a)	1
2	(a)	1
3	(c)	1
4	(d)	1
5	(a)	1
6	(b)	1
7	(b)	1
8	(d)	1
9	(a)	1
10	(b)	1
11	(c)	1
12	(d)	1
13	(a)	1
14	(d)	1
15	(c)	1
16	(d)	1
17	Section-B (i) We know that $v_d = \frac{eV\tau}{ml} \propto \frac{1}{l}$ When length is tripled, the drift velocity becomes one-third. (ii) $R = \rho \frac{l}{A}$, $l' = 3l$ New resistance $R' = \rho \frac{l'}{A'} = \rho \times \frac{3l}{A/3} = 9R \Rightarrow R' = 9R$ Hence, the new resistance will be 9 times the original.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
18	Descending order of wavelength Microwaves < Visible light < Ultraviolet radiation < γ-rays Any one use of each Microwave, γ-rays	1 $\frac{1}{2} + \frac{1}{2}$
19	Correct ray diagram (i) Correct ray diagram (ii)	1 1
20	Correct graph (i) Correct graph (ii)	1 1
21	Labeled diagram of p-n junction Explanation of formation depletion region and potential barrier	1 1
21 OR	OR Circuit diagram of full wave rectifier Input and output wave form	1 1

	Section-C	
22	Derivation of force/length Definition of one ampere B_1 acts perpendicular and into the plane of the diagram by Right Hand Thumb Rule. So, the angle between I and B_1 is 90°. l is length of the conductor. $F_{21} = \mu_0 I_1 I_2 l / 2\pi r$ $B_2 = \mu_0 I_2 / 2\pi r$ Magnetic Field on PQ due to current in RS is Force acting on PQ due to current I_2 through it is $F_{12} = \frac{\mu_0 I_2}{2\pi r} I_1 l \sin 90^\circ \quad \text{or} \quad F_{12} = \frac{\mu_0 I_1 I_2 l}{2\pi r}$ $F_{12} = F_{21} = F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$ Force per unit length of the conductor is $F/l = \frac{\mu_0 I_1 I_2}{2\pi r} \quad \text{N/m}$ By Fleming's Left Hand Rule, the conductors experience force towards each other and hence attract each other. By Fleming's Left Hand Rule, the conductors experience force away from each other and hence repel each other. Definition of Ampere: Force per unit length of the conductor is $F/l = \frac{\mu_0 I_1 I_2}{2\pi r} \quad \text{N/m}$ When $I_1 = I_2 = 1$ Ampere and $r = 1$ m, then $F = 2 \times 10^{-7}$ N/m. One ampere is that current which, if passed in each of two parallel conductors of infinite length and placed 1 m apart in vacuum causes each conductor to experience a force of 2×10^{-7} Newton per metre of length of the conductor.	2 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1
	OR	
	a) Correct Diagram & Principle working b) function of radial MF and soft iron core	$\frac{1}{2}, \frac{1}{2}$ 1 1
23	Binding energy curve Two conclusions	1 1+1

24	(i) Two advantages (ii) Low cost (iii) Correct explanation OR Correct explanation	1 1 1
28	The energy E of a photon of wavelength 275 nm is given by $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{275 \times 10^{-9}} \text{ J}$ $= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{275 \times 10^{-9} \times 1.6 \times 10^{-19}} \text{ eV}$ $= 4.5 \text{ eV}$ Therefore, transition B corresponds to the emission of photon of $\lambda = 275 \text{ nm}$. for which energy change $= 0 - (-4.5) = 4.5 \text{ eV}$ Energy of emitted photon, $E \propto 1/\lambda$ (i) For transition A, the energy emission E is minimum, corresponds to the emission of radiation of maximum wavelength. (ii) For transition D, the energy emission E is maximum, corresponds to the emission of radiation of minimum wavelength.	1 1/2 1/2 1/2 1/2
29	(i) (a) (ii) (b) (iii) (c) (iv) (d) $M.P. = -\frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$	1 1 1 1
30	(i) (a) (ii) (b) (iii) (c) (iv) (d)	1 1 1 1
Section-D		
31(a)	Diagram of ac generator & Principle Working of ac generator Working: As the armature coil $ABCD$ rotates, the magnetic flux linked with it changes. Hence, an emf is induced in the coil and current flows in it.  (i) The variation of magnetic flux with time is given by $\phi = NBA \cos \omega t$	1 2 1/2

	(ii) The variation of alternating emf with time is given by  $\Rightarrow \frac{d\phi}{dt} = -BA\omega \sin \omega t$ $\Rightarrow \epsilon = \frac{-Nd\phi}{dt}$ $= NBA\omega \sin \omega t$ $\Rightarrow \epsilon = -\epsilon_0 \sin \omega t$	1/2 1/2 1/2
(b)	$N = 100, B = 0.8 \text{ T}, R = 100 \Omega$ $A = 0.5 \text{ m}^2, \omega = 60 \text{ rad/s}$ Maximum emf $= \epsilon_0 = NBA\omega = 100 \times 0.8 \times 0.5 \times 60 = 2400 \text{ V}$ Power dissipated, $P = \frac{\epsilon^2}{R}$ $\Rightarrow P = \frac{\epsilon_0^2}{2R} \quad (\epsilon = \epsilon_0/\sqrt{2})$ $= \frac{2400 \times 2400}{2 \times 100}$ $= 28800 \text{ W} = 28.8 \text{ kW}$	1 1
	OR	
	(a) At Resonance $X_L = X_C$, $\omega_o L = 1/\omega_o C$, $\omega_o = 1/\sqrt{LC}$ $\omega_o = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = 50 \text{ rev/s}$ (b) Impedance at resonance $z = R, Z = 40 \Omega$ Current amplitude $I_o = \frac{E_{rms} \times \sqrt{2}}{Z} = \frac{230 \times \sqrt{2}}{40} = 8.1 \text{ A}$ (c) $I_{rms} = \frac{E_{rms}}{Z} = \frac{230}{40} = \frac{23}{4} \text{ A}$ $V_L = (I_{rms} \times \omega_o L) = \frac{23}{4} \times 50 \times 5 \text{ V} = 1437.5 \text{ V}$ $V_R = (I_{rms} \times R) = \frac{23}{4} \times 40 \text{ V} = 230 \text{ V}$ (d) Power factor, $\cos \phi = 1$ (for Pure resistive circuit $\phi = 0$) (e) 	1 1/2, 1/2 1 1 1 1

32	(a) Statement of gauss law & Fig. Derivation (i) E.F. inside sphere (ii) E.F. outside (b) Flux through $S_1(\phi_1) = \frac{Q}{\epsilon_0} \dots(i)$ Flux through $S_2(\phi_2) = \frac{Q+2Q}{\epsilon_0} = \frac{3Q}{\epsilon_0} \dots(ii)$ $\therefore$ Ratio of flux = $\frac{\phi_1}{\phi_2} = \frac{Q/\epsilon_0}{3Q/\epsilon_0} = \frac{1}{3}$	1 $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ 1
	OR	
	(a) Fig. Torque experienced by an Electric Dipole in a Uniform Electric Field As the forces $q\vec{E}$ and $-q\vec{E}$ act at two different points, they will produce a torque. Torque = either of the force $\times$ perpendicular distance between them $\therefore \tau = qE \times 2l \sin \theta = q 2l E \sin \theta = pE \sin \theta$ Vectorially, we can express as $\vec{\tau} = \vec{p} \times \vec{E} = pE \sin \theta \hat{n}$  Net force $-qE + qE = 0$	$\frac{1}{2}$ 1 $\frac{1}{2}$
	(b) We know, $\tau = pE \sin \theta$ $\Rightarrow 8\sqrt{3} = p \times 10^5 \times \sin 60^\circ$ $\Rightarrow 8\sqrt{3} = p \times 10^5 \times \frac{\sqrt{3}}{2}$ $\Rightarrow p = 16 \times 10^{-5}$ $= 1.6 \times 10^{-4} \text{ Cm}$ Also, $p = q(2L)$ $\Rightarrow 1.6 \times 10^{-4} = q \times 2 \times 10^{-2}$ $\Rightarrow q = \frac{1.6 \times 10^{-4}}{2 \times 10^{-2}}$ $q = 0.8 \times 10^{-2} = 8 \times 10^{-3} \text{ C}$ $q = 8 \text{ mC}$ $\therefore$ Magnitude of the charge on the dipole is 8 mC Now, potential energy of dipole, $U = -pE \cos \theta$ $= -1.6 \times 10^{-4} \times 10^5 \times \cos 60^\circ$ $= -1.6 \times 10 \times \frac{1}{2} = -8 \text{ J}$ $\therefore$ Potential energy of the dipole is -8 J $\therefore$ $E = 10^5 \text{ N/C}$ $\theta = 60^\circ$ $\tau = 8\sqrt{3} \text{ Nm}$ where p is dipole moment	1 1 1 1
33	(a) Ray diagram Derivation of relation Any Two factor	1 1 1
	(b)	1

	As $\mu = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}} \therefore \sqrt{3} = \frac{\sin 60^\circ}{\sin 30^\circ}$ or $\sin \frac{60^\circ + \delta_m}{2} = \sqrt{3} \times \frac{1}{2} = \sin 60^\circ$ or $\frac{\delta_m + 60^\circ}{2} = 60^\circ$ or $\delta_m = 120 - 60 = 60^\circ$	$\frac{1}{2}$ $\frac{1}{2}$
	OR	
33	(a) Statement of Huygens principle Diagram showing reflected wavefront Verification of law (b) Finding distance of object from mirror	1 1 1 2
	(a) Huygen's principle Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets emanating from the wavefront are usually referred to as secondary wavelets, a common tangent to all these spheres gives the new position of the wavefront at a later time.  Verification of law of reflection In $\triangle AEC$ & $\triangle CBA$ $EC = AB$ (c x t each) $\angle AEC = \angle CBA$ (90° each) $AC = AC$ (common side) By RHS congruency $\triangle AEC \cong \triangle CBA$ $\Rightarrow \angle i = \angle r$ (b) $m = +3$, $f = -12$ cm, $u = ?$ $m = -\frac{v}{u} = 3 \Rightarrow v = -3u$ using mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\frac{1}{-3u} + \frac{1}{u} = \frac{1}{-12}$ $u = -8$ cm	